

● EASY

● ALGEBRA

Linear equations in two variables

03

10 questions · ~15 min

Q1

GRID-IN ALGEBRA

Vivian bought party hats and cupcakes for \$ 71. Each package of party hats cost \$ 3, and each cupcake cost \$ 1. If Vivian bought 10 packages of party hats, how many cupcakes did she buy?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Last week, an interior designer earned a total of \$ 1,258 from consulting for x hours and drawing up plans for y hours. The equation $68x + 85y = 1,258$ represents this situation. Which of the following is the best interpretation of 68 in this context?

- A. The interior designer earned \$ 68 per hour consulting last week.
- B. The interior designer worked 68 hours drawing up plans last week.
- C. The interior designer earned \$ 68 per hour drawing up plans last week.
- D. The interior designer worked 68 hours consulting last week.

A gardener buys two kinds of fertilizer. Fertilizer A contains 60% filler materials by weight and Fertilizer B contains 40% filler materials by weight. Together, the fertilizers bought by the gardener contain a total of 240 pounds of filler materials. Which equation models this relationship, where x is the number of pounds of Fertilizer A and y is the number of pounds of Fertilizer B?

- A. $0.4x + 0.6y = 240$
- B. $0.6x + 0.4y = 240$
- C. $40x + 60y = 240$
- D. $60x + 40y = 240$

Q4

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A producer is creating a video with a length of 70 minutes. The video will consist of segments that are 1 minute long and segments that are 3 minutes long. Which equation represents this situation, where x represents the number of 1-minute segments and y represents the number of 3-minute segments?

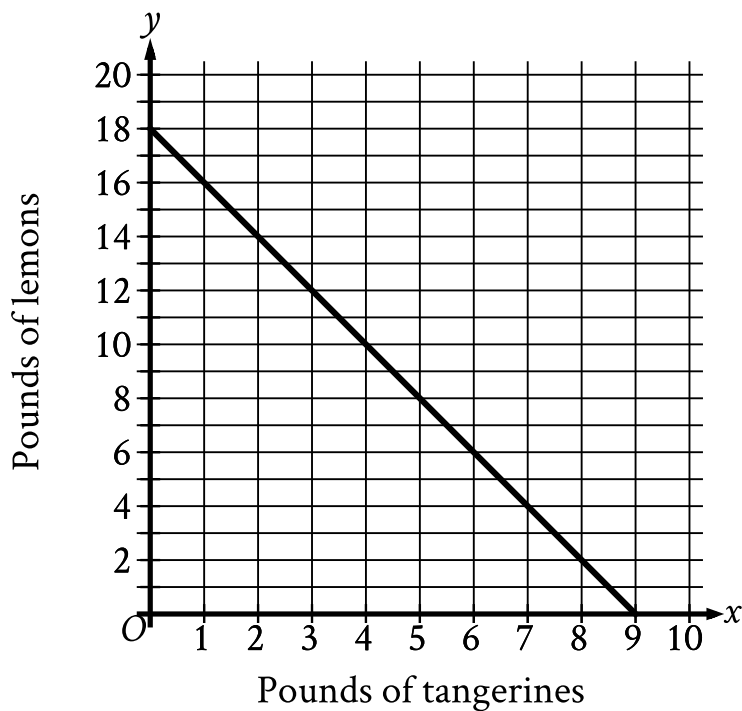
- A. $4xy = 70$
- B. $4x + y = 70$
- C. $3x + y = 70$
- D. $x + 3y = 70$

Q5

ALGEBRA

In a chess tournament, each participant earns 1 point for each game the participant plays that ends in a draw and 3 points for each game the participant wins. A certain participant in this tournament has earned 41 points. Which equation represents this situation, where d represents the number of games this participant has played that ended in a draw and w represents the number of games this participant has won?

- A. $d + 3w = 41$
- B. $3d + w = 41$
- C. $d + \frac{w}{3} = 41$
- D. $\frac{d}{3} + w = 41$



- The line slants sharply down from left to right.
- The line passes through the following points:
 - (0 comma 18)
 - (4 comma 10)
 - (7 comma 4)
 - (9 comma 0)

The graph shows the possible combinations of the number of pounds of tangerines and lemons that could be purchased for \$ 18 at a certain store. If Melvin purchased lemons and 4 pounds of tangerines for a total of \$ 18, how many pounds of lemons did he purchase?

- A. 7
- B. 10

C. 14

D. 16

$$y = x + 4$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
0	4
1	5
2	6

B.

x	y
0	6
1	5
2	4

C.

x	y
0	2
1	1
2	0

D.

x	y
0	0
1	1
2	2

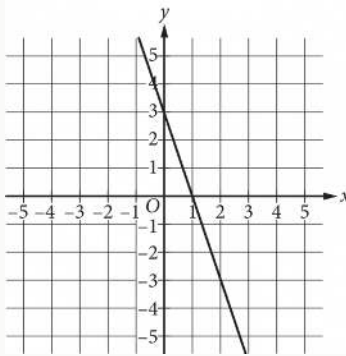
Q8

ALGEBRA

$$x + y = 75$$

The equation above relates the number of minutes, x , Maria spends running each day and the number of minutes, y , she spends biking each day. In the equation, what does the number 75 represent?

- A. The number of minutes spent running each day
- B. The number of minutes spent biking each day
- C. The total number of minutes spent running and biking each day
- D. The number of minutes spent biking for each minute spent running



What is the equation of the line shown in the xy -plane above?

- A. $y = 3x - 3$
- B. $y = -3x + 3$
- C. $y = \frac{1}{3}x - 3$
- D. $y = -\frac{1}{3}x + 3$

A city's total expense budget for one year was x million dollars. The city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. Which of the following represents the relationship between x and y in this context?

- A. $x + y = 201$
- B. $x - y = 201$
- C. $2x - y = 201$
- D. $y - x = 201$